

Tue, Nov 4 at 4:02 PM

35-year-old client with Multiple Sclerosis who is interested in trialling the Bioness to assist with her foot drop. Her current assessment findings are below:

Right Sided Strength: 0/5 Ankle Dorsiflexion, 3/5 Ankle Plantarflexion, 0/5 Ankle Eversion, 1/5 Ankle Inversion, 4/5 Knee extensor/flexor strength, 5/5 Hip abd/ext strength

Ankle DF ROM: Knee Ext: ~ -3 degrees.

Moderate spasticity in soleus and gastrocs, with gastrocs > soleus, mild spasticity in ankle invertors

Other things to note: Right sided toe curling becomes evident with prolonged standing/walking.

Mobility: Independent with a 4-Wheeled walker indoors.

Gait: Right Knee hyperextension on stance, right foot drop and inversion in mid-swing.

She has trialled an off the shelf AFO (OAPL Supra-Lite AFO), however this has been unsuccessful and ill-fitting and we are in the process of sourcing a custom made AFO from KOrthotics. However at present, she does not have funding for Orthotics in her NDIS plan.

This is an application for M for NDIS-funded community trial of the Bioness L300 Go Functional Electrical Stimulation (FES) system. M has a confirmed diagnosis of Multiple Sclerosis (MS) and presents with significant foot drop and gait impairment on her right side, which impacts mobility, community participation, and safety.

M demonstrates reduced dorsiflexion during swing phase, compromised walking efficiency, increased risk of falls, and reduced walking tolerance. These impairments substantially limit independence in daily and community activities. M has trialled off the shelf AFO with her physiotherapist, which has been unsuccessful and ill-fitting likely due to the presence of contractures with spasticity. Also, she did not like bulkiness of the AFO.

M attended Korthotics on 24th of September 2025 with her mother and physiotherapist to try Bioness to determine eligibility of Functional Electrical Stimulation (FES). The Bioness System offers both orthotic (immediate) and therapeutic (long-term) benefits by stimulating the common peroneal nerve to assist dorsiflexion during gait. Unlike a passive orthosis, FES promotes more natural walking patterns, supports muscle activity, and may reduce secondary complications associated with reduced mobility.

M had short time trial of Bioness in our clinic on her right side that have increased foot clearance in swing phase and an increase stability in stance phase. Also, increased her walking speed and reduced fatigue (walked 5 times more than normal).

A community trial is necessary to assess its effectiveness in real-life environments, as clinical trial alone cannot reflect day-to-day demands. This trial ensures that NDIS investment in high-cost technology is supported by functional evidence.

Here's a polished and professionally formatted NDIS support letter you can use or adapt for submission. It presents your clinical findings and reasoning clearly and persuasively for NDIS review.

NDIS Support Letter – Application for Community Trial of Bioness L300 Go (FES System)

Client: M [Surname]

Date of Birth: [DD/MM/YYYY]

Diagnosis: Multiple Sclerosis (MS)

Date: [Insert Date]

Therapist: [Your Name, Title, Registration Number]

Purpose of Report

This report supports an application for NDIS-funded community trial of the Bioness L300 Go Functional Electrical Stimulation (FES) system for M, a 35-year-old woman diagnosed with Multiple Sclerosis. The aim of the trial is to determine the effectiveness of FES in improving M's mobility, walking safety, and community participation.

Clinical Presentation

Right Lower Limb Strength:

Ankle Dorsiflexion: 0/5

Ankle Plantarflexion: 3/5

Ankle Eversion: 0/5

Ankle Inversion: 1/5

Knee Flexion/Extension: 4/5

Hip Abduction/Extension: 5/5

Range of Motion (ROM):

Right Ankle Dorsiflexion (Knee Extended): $\sim -3^\circ$

Tone:

Moderate spasticity in soleus and gastrocnemius (gastrocs > soleus)

Mild spasticity in ankle invertors

Additional Findings:

Right-sided toe curling becomes evident with prolonged standing or walking.

Gait:

Right knee hyperextension in stance phase

Right foot drop and inversion during mid-swing phase

Mobility:

Independent ambulation with a 4-wheeled walker indoors

Interventions Trialled

M has trialled an off-the-shelf OAPL Supra-Lite AFO, which was unsuccessful and ill-fitting. The AFO was found unsuitable due to the combination of spasticity, mild contracture, and M's sensory preferences (she reported discomfort and disliked the bulkiness). A custom AFO is being considered through KOrthotics; however, there is currently no orthotic funding available in her NDIS plan.

Rationale for FES (Bioness L300 Go)

M attended an appointment at KOrthotics on 24 September 2025 with her mother and physiotherapist for a clinical trial of the Bioness L300 Go. The device applies electrical stimulation to the common peroneal nerve, producing ankle dorsiflexion during gait. This directly addresses foot drop and facilitates a more natural and efficient walking pattern.

The Bioness system provides both orthotic (immediate) and therapeutic (long-term) benefits:

Promotes active muscle engagement

Supports a more natural gait cycle

May reduce spasticity and prevent secondary complications

Enhances mobility, independence, and community participation

Clinical Trial Outcomes

During a brief in-clinic trial, M demonstrated:

Improved foot clearance in swing phase

Increased stability in stance phase

Greater walking speed and reduced fatigue

Reported walking approximately five times further than her baseline capacity

These results indicate that FES may significantly enhance M's functional mobility and quality of life.

Justification for NDIS-Funded Community Trial

A community-based trial is recommended to evaluate:

The effectiveness of the device in day-to-day environments

Impact on safety, fatigue, and mobility over time

Consistency of benefit across varied terrain and activities

A trial is essential to determine whether ongoing provision of FES represents value for money and meets the NDIS reasonable and necessary criteria before a full purchase is considered.

Summary and Recommendation

Given M's:

Persistent right-sided foot drop and gait instability due to MS

Ineffectiveness of AFO intervention

Positive response to initial FES trial

...it is clinically recommended that the NDIS fund a community trial of the Bioness L300 Go FES system to assess suitability and functional outcomes in real-world conditions.

Therapist Name: [Full Name]

Position/Role: [e.g., Physiotherapist / Orthotist]

Organisation: [Organisation Name]

Registration No.: [AHPRA / Membership No.]

Contact: [Phone / Email]

Date: [Insert Date]

Would you like me to adapt this into an orthotist's version (written from their perspective rather than physiotherapist), or keep it as a physiotherapist support letter?

Using above details to write NDIS support letter for community trial written by orthotist

Here's a professionally written NDIS support letter for a community trial of the Bioness L300 Go, tailored to be authored by an orthotist. It uses all your provided clinical details and presents them clearly and persuasively in the tone and structure NDIS expects.

[Orthotist Letterhead]

Date: [Insert Date]

To: National Disability Insurance Agency (NDIA)

Re: Application for NDIS-Funded Community Trial – Bioness L300 Go (Functional Electrical Stimulation System)

Participant: M [Surname]

Date of Birth: [DD/MM/YYYY]

Diagnosis: Multiple Sclerosis (MS)

Referring Therapist: [Physiotherapist Name, if applicable]

Orthotist: [Your Name], Certified Orthotist/Prosthetist

Purpose of Report

This letter is written in support of funding a community trial of the Bioness L300 Go Functional Electrical Stimulation (FES) system for M, a 35-year-old woman with Multiple Sclerosis. M presents with significant right-sided foot drop and gait deviations, which impact her safety, mobility, and independence in daily and community activities.

Clinical Presentation

During assessment, M demonstrated the following findings:

Right Lower Limb Strength:

Ankle Dorsiflexion: 0/5

Ankle Plantarflexion: 3/5

Ankle Eversion: 0/5

Ankle Inversion: 1/5

Knee Flexion/Extension: 4/5

Hip Abduction/Extension: 5/5

Range of Motion (Right Ankle):

Dorsiflexion with knee extended: approximately -3°

Tone:

Moderate spasticity in soleus and gastrocnemius (gastrocnemius > soleus)

Mild spasticity in ankle invertors

Additional Observations:

Right-sided toe curling evident with prolonged standing or walking

Mobility and Gait:

Independent with a 4-wheeled walker indoors

Exhibits right knee hyperextension in stance phase

Foot drop and inversion during mid-swing phase

Previous Orthotic Intervention

M trialled an off-the-shelf OAPL Supra-Lite AFO, which proved unsuccessful and ill-fitting due to the combination of spasticity, mild contracture, and her limb shape. M also reported discomfort and difficulty tolerating the AFO due to its bulkiness.

A custom-made AFO is currently being considered through KOrthotics; however, orthotic funding is not presently included in M's NDIS plan.

Trial of Functional Electrical Stimulation (FES)

M attended KOrthotics on 24 September 2025 with her physiotherapist and mother for an in-clinic trial of the Bioness L300 Go system. The device applies targeted stimulation to the common peroneal nerve, producing ankle dorsiflexion during the swing phase and assisting in achieving a more natural gait pattern.

During the short in-clinic trial, M demonstrated:

Noticeable improvement in foot clearance during swing

Improved stability in stance phase

Increased walking speed and distance (approximately five times her baseline walking distance)

Reduced fatigue and improved walking confidence

These immediate orthotic effects indicate that FES may be an appropriate and effective intervention for M's presentation.

Clinical Rationale

The Bioness L300 Go offers both orthotic (immediate) and therapeutic (long-term) benefits:

Actively facilitates dorsiflexion and eversion during gait

Encourages muscle activation rather than passive positioning

Supports a more natural gait cycle

May assist in reducing secondary complications associated with immobility (e.g., contractures, disuse weakness)

Enhances safety, walking efficiency, and community participation

Given that M has not tolerated conventional orthoses successfully, FES presents a functionally suitable and potentially superior alternative to traditional AFO management.

Recommendation

It is recommended that the NDIS fund a community-based trial of the Bioness L300 Go FES system for M.

A community trial will:

Allow evaluation of the device's real-world functional impact on walking and fatigue management

Determine its long-term suitability for her daily and community activities

Ensure that any future device funding is clinically justified and cost-effective

Summary

M's clinical presentation of right-sided foot drop and gait instability secondary to Multiple Sclerosis has not been successfully managed with conventional orthoses. Her positive response during the in-clinic FES trial suggests that the Bioness L300 Go may significantly enhance her mobility, safety, and participation.

A community trial is essential to confirm these benefits in her everyday environments before proceeding with a full prescription.

Kind regards,

[Your Full Name]

Certified Orthotist/Prosthetist

[Clinic/Organisation Name]

[Contact Number / Email]

AOPA Membership / Registration No.: [Insert]

Would you like me to include a research evidence paragraph (summarising key studies supporting FES use in MS foot drop) to strengthen the letter for NDIS reviewers? It can make the justification even more compelling.

Make into word file

The NDIS support letter has been formatted and saved as a Word document. You can download it using the link below:



NDIS_Support_Letter_Bioness_L300Go_Orthotist.docx